UNIVERSITY OF TORONTO

ECE496 PROPOSAL

TEAM 2025162, SECTION 2

Vision-Guided Automated Foosball Opponent

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October 18th, 2025

Executive Summary

This project seeks to develop a vision-guided automated foosball opponent, combining real-time computerized game state tracking and mechanical actuation of rods. Foosball is a tabletop game with similarities to soccer, where two players can control small players attached to rods which can be moved side to side and rotated to kick the ball. This is following a trend of projects that re-imagine casual activities such as darts and chess through robotics and real-time tracking.

Our primary motivation is due to the potential to involve multiple concepts of Electrical and Computer Engineering, providing a diverse coverage of key concepts learned by the team throughout their undergraduate education. It is important in practice as computerized opponents in other popular games such as chess or go have gained recognition, but foosball has not yet been implemented in a reproducible way with off-the-shelf components and techniques.

The final product will be a removable attachment to a full-size foosball table (standard table dimensions used in professional foosball competitions) which can perform defensive gameplay for at least two rods for one side, with a reaction time around 0.2 seconds; however, effective offensive gameplay is an additional goal if these objectives are completed.

For effective tracking, simple computer vision techniques (e.g. color filtering, edge detection) will take advantage of visual input devices able to take fully synchronized frames at above 60 FPS. This must be combined with controller hardware whose calculations are able to keep up with framerate. A universal interface will allow viewing of game state and visually represented calculations while in implementation different hardware will be used to focus on control only. Off-the-shelf motors providing at least 1.158Nm of torque and 1000RPM of speed will ensure realistic reactions in gameplay, with one set for rotational movement and another for translational movement of the rods. The aim is to have the total cost of all components stay under $500 per member of the team.

This project seeks to transform foosball into a demonstration of Electrical and Computer Engineering concepts and allow players to engage and practice in an effective single-player mode, gathering interest in both the engineering and the game.

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1.0 Introduction

Foosball has long been a popular tabletop game enjoyed around the world in family homes, offices, recreational centres, and professional settings. The appeal lies in its fast-paced simulated soccer gameplay and its unique blend of strategy with luck. Traditionally, this tabletop game has always been played between humans, with the winner determined on the basis of their reflexes, positioning, and ball control. However, with the recent advancements of robotics, it has become increasingly feasible to bring autonomy into various interactive games such as foosball.

Autonomous opponents in the game of foosball highlights multiple levels of robotics research challenges, including but not limited to decision-making, visual perception of fast-moving objects, and prediction-based actuation; all of which mirror modern day complex robotic designs such as autonomous vehicles or automated assembly lines. In addition, the development of an automated foosball opponent serves both educational and research value. A fully integrated control system with the foosball table would demonstrate applied robotics concepts in real time; serving as a practical device to further explore human-machine interfaces and advanced interactive robotics.

2.0 Background and Motivation

In foosball, two players control figures mounted on metal rods, moving in a linear and rotational fashion to simulate a game of soccer. Like other casual activities, such as darts and chess, it is highly accessible and holds potential as a platform for technological enhancement. Recently, there has been a growing interest in re-imagining how these games can be played through automation and real-time control. For example, a viral video by Mark Rober demonstrates a dart board that automatically moves to intercept the player's darts [1]. Foosball is a similar candidate for this type of enhancement: its core mechanics are simple, yet the challenge of automation involves an interesting mix of hardware and software development, ultimately resulting in an engaging platform to demonstrate concepts in engineering.

Commercial solutions that automate tabletop games do exist, but focus on smaller formats - such as Chessnut, a chessboard that moves the pieces by itself. Foosball's larger scale has historically limited solutions to hobbyist or student projects. Some studies have focused on individual subsystems of an automated system, such as a paper from Eindhoven University that discusses methods of ball tracking in a ”semi-automated foosball table” [2]. A few more complete projects have been produced by student groups over the years, including one from California Polytechnic University, in which a system with full movement for four rods was built [3]. This group was able to implement a sophisticated mechanical setup but was unable to finish a working vision system. Student projects often fall into two categories: a lack of proper funding results in an underpowered system, or the use of expensive sponsored components results in a successful system limited in its reproducibility. Our project seeks to demonstrate an automated system using off-the-shelf components, balancing performance and an accessible design process. We aim to produce a design that can be referenced and expanded on in the future.

This project also presents an opportunity to build a practical capstone design that integrates skills from multiple fields of electrical and computer engineering in a unified system. Automation of a football table requires knowledge of power, embedded systems, real-time communication, computer vision, and algorithm design. Other students will also be able to interact with our design, and future groups can take inspiration from and build on our open design process, lowering the barrier to entry in terms of funding and complexity. Ultimately, this project highlights how familiar games can platform educational concepts and engineering tools.

3.0 Problem Statement

Traditional foosball tables rely entirely on human players to control the rods and react to the ball's movement. While foosball is a popular recreational activity, current tables offer no digital aspect, let alone an automated opponent that can match human reaction times and conduct strategic play, and may also be difficult for beginners to get into the game, caused by the differences in player skill levels. This limits the ability to use foosball as a platform for exploring human-machine interaction and also forces foosball to an often unbalanced two-people play.

Therefore, there is a need for an autonomous foosball opponent, guided by a computer through visional devices such as a camera to integrate real-time ball tracking and mechanical actuation of rods. Such a system would not only provide an engaging opponent for casual play, but also provide a unique opportunity to showcase engineering skills, including computer vision, embedded systems, and software-hardware integration, in a recreational setting that is both accessible and engaging to a wide audience.

To achieve this, we will start with working on the control of two rods that can coordinate with each other, then potentially implementing all four rods on a standard sized foosball table. The development will be divided into four subsystems: the visual unit, calculation unit, control unit, and the power unit, with each team member primarily responsible for one area.

4.0 Project Goal

The goal of this project is to develop a control system that automates one side of a foosball table's rods, allowing for primarily reactive but potentially active play, controlling the ball at upwards of 1m/s and defending up to one minute of continuous play.

5.0 System Requirements Specification

The following section details our system requirements for our project, with the specifications split into a vision and strategy section and a mechatronics section.

5.1 Vision and Strategy

Table 1 shows the systems requirements for the vision and strategy aspects of our design requirements for the foosball table.

Table 1. Requirements, constraints, and objectives of the vision/strategy portion.

ID	Specification	Description
1	Input device framerate > 60 FPS.	Objective: Higher is better. This is from a testimonial by a representative of Foosball Canada, who has anecdotally mentioned that conventional 60 FPS webcams (or similar input devices) are insufficient to capture all movement. Off-the-shelf offerings exceeding this minimum vary but often come with 100 FPS or higher.
2	Input device must detect in full color.	Requirement: For more effective detection of objects by color.
3	Calculation of required rod movement including lateral position and kick trigger for all motorized rods in $< \frac{1}{input framerate}$ seconds.	Requirement: Each frame should have its own calculation to update the rod position with the latest information.
4	Input device must take all areas of frame simultaneously.	Requirement: Each part of the frame should be equally accurate to the time the information was processed.
5	Software must have a visual ”debug” mode which shows detections, activations, and ball motion state.	Requirement: The automated system's vision processing status should have a visible mode for finetuning purposes that can be switched off for optimization when running in normal use.
6	Device must be able to detect whole ball and player positions in each frame	Requirement: Position of ball is required for defensive actions, and player positions for strategic offensive actions.
7	Input device must not be visually obstructed on the way to the ball or the players	Requirement: Cannot see game state if game state is visually obstructed.
8	Input device must connect using a universal interface (e.g. USB).	Constraint: Control hardware will change between simulation and action, and a common interface should be used for the software to use the same structure of input.

5.2 Mechatronics

Table 2 shows the systems requirements for the mechatronic aspects of our design requirements for the foosball table.

Table 2. Requirements, constraints, and objectives of the mechatronics portion.

ID	Specification	Description
9	Motor Input Power: Minimum 24VDC and 7.5A	Requirement: Power supply must meet this specification. See Appendix B for further details.
10	Components Price: Does not exceed an investment of $500 from each group member, we must assume there will not be any re-reimbursements.	Constraint: Total cost cannot exceed this amount, therefore we will not acquire top-of-the-line components.
11	Construction: System does not make permanent modifications to the foosball table, rather acts as an accessory that can be removed.	Objective: The original foosball table should remain intact and usable.
12	Rotational Motor Specifications: The motor must rotate at least 120 degrees, and output peak torque and speed of 1.158Nm and 1000RPM.	Requirement: This is for realistic gameplay. See Appendix C for detailed calculations.
13	Translational Motor Specifications: The motor must support up to 2kg in translational movement, rotate 360 degrees, and output a peak torque and speed of at least 1.2Nm and 500RPM.	Requirement: This is for realistic gameplay. See Appendix D for detailed calculations.
14	Linear Guide Rail: The maximum travel length is 400mm (measured internally by the team).	Requirement: The travel distance of the linear guide rail must be within 400mm, however, rods with shorter travel can use proportionally shorter guide rails.
15	Reaction Time: The reaction time of the motor together should average at 0.2s.	Objective: An average human reaction time is around 0.25s [4], therefore, the estimated motor reaction time needs to be faster to account for the computer processing time for our visual perception setup.
16	Automated Rods: Two or more rods.	Objective: Setup will have two or more automated rods
17	Defensive Ability: One or more minutes.	Objective: The system will be able to defend against a goal for at least one minute against an untrained player.
18	Does Not Obstruct Human Side: No interference.	Requirement: Does not interfere with the opposing side of the foosball table or the abilities of the human to play.

6.0 System Block Diagram

Figure 1 shows the overall block diagram for our system, separated into five individual units. The labels on the left denote the initial separation of responsibilities of the group members.

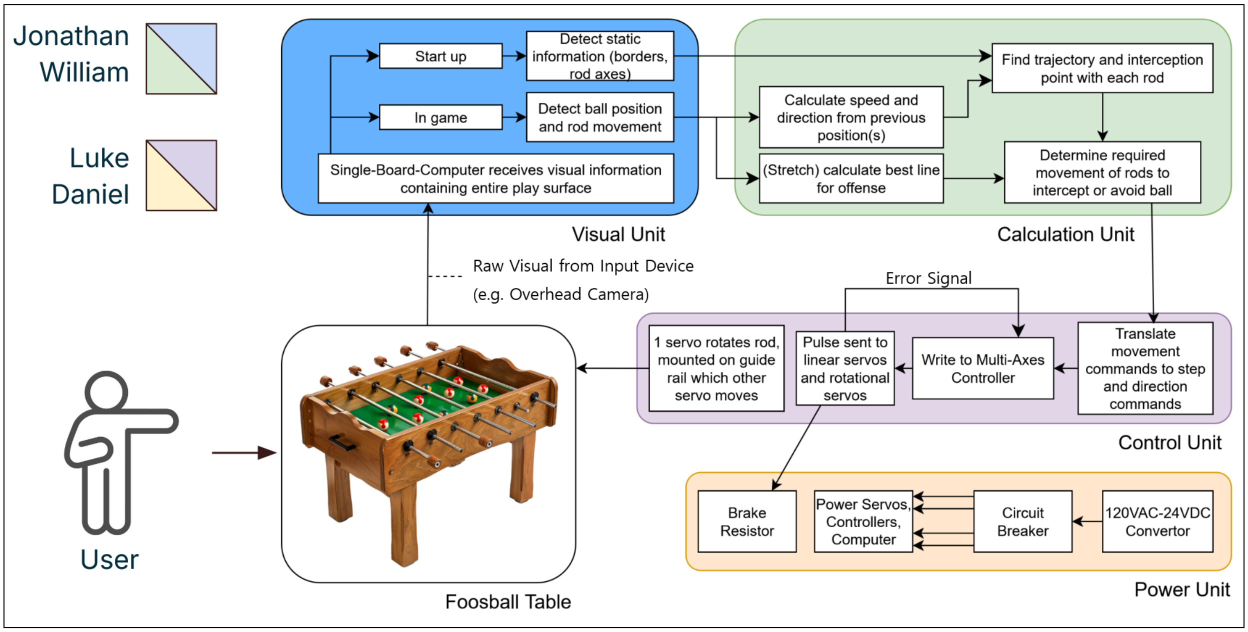


Figure 1. Block diagram of the automated foosball opponent system.

Starting with the foosball table, which the user interacts with on one side, an overhead video camera feeds raw footage into the ”Visual Unit”. This unit is responsible for processing this information and generating positioning for the elements of the game, including the field boundaries, rod axes, and most importantly ball movement. This data is fed into the ”Calculation Unit”, a micro-controller that will determine the required positioning of the automated rods to intercept the ball. Finally, the ”Control Unit” translates this into commands output to the motors, which move and rotate the rods. Behind this flow, the ”Power Unit” is necessary for supplying all of these different elements with sufficient power, including the motors and controller. All of these units must work together within a short time-frame to achieve real-time play.

7.0 Risk Assessment

The following section identifies four risks that could negatively affect the outcome of our project.

7.1 Uneven and Inconsistent Lighting

- During some experiments with the prototype, it was noted that edges were not as visible or colors may differ depending on lighting angle and intensity, as well as the general temperature of any light source.
- These effects may impact detection of the ball and rod, unable to either capture the full area (thus making the detected position off-center) or not capture at all, which makes accurate calculations at the

desired framerate impossible.

- This is likely to happen given the variable amounts of outside sunlight and the fluctuation in artificial lighting depending on setting. To mitigate this problem, a method to evenly light the table (not affected by the shape of any light source, and thus most likely diffused) in a manner not invasive to gameplay is necessary.

7.2 Insufficient Accuracy of Rod Detection

- One concern about any potential methods of vision detection is that the rods may be partially or wholly undetected. This leads to problems in determining the position of foos-figures to analyze any openings to shoot the ball, either inaccurate or not knowing the positions at all. The majority of cases are caused by the aforementioned lighting issue which can be mitigated, and thus this is not a major problem for the defensive case which only has to intercept the ball and shoot it in any direction away from the motorized team's goal (position of defense rods are known and there is the width of each foos-figure as a margin of error in interception).
- However, in the offensive case, the angle of attack (and the consequential line taken by the ball, in comparison with the one which minimizes interceptions by the other team) will be dramatically altered by small differences in where the ball is struck.
- While this is likely to happen, coinciding with the lighting issue, since offensive play is a stretch goal on its own, it is not highly urgent. This can be mitigated by creating backup methods of lateral rod position detection for non-motorized rods, or at the very least accurate estimates if the disparity is only a few frames long.

7.3 Motor Power is Insufficient

- A key component in the functionality of the automated system is the ability of the selected motors to provide enough power to rotate and move the rods so that the ball can be intercepted and kicked. The probability this will occur is low for the motors responsible for rotation, as the selected motors were above the calculated requirements. The motors guiding the linear system, however, are more at risk due to the more complex setup and weight they have to move.
- The consequence of one of the motors failing in this regard would be a limitation on the overall system to respond to a user's inputs in a realistic way. The game would not be playable as the rods would likely miss interception, being too slow to react, or not able to kick the ball with enough force to move it.
- The root cause of this risk is both the motors themselves and our chosen setup. If either motor is too weak, it can be replaced with a more powerful version. However, this can only go so far in our goal to build an inexpensive and reproducible system. In addition, the stronger and heavier the rotational motor is the more weight the linear motor has to move. Therefore, the specifications can be modified to use a less powerful but lighter motor for rotational movement. This means kicking the ball with less

force, but the system would at least be able to intercept it. Ultimately, if no combination of motors is able to both intercept the ball and deliver enough force to move it, the scope of the project must then be restricted to only gameplay in which the user plays at a slow speed. This would not be realistic gameplay, but would still allow for a demonstration of potential capabilities.

7.4 Processing Time is Too Slow

- Another possibile failure point is that the commands cannot be generated in time to feed to the motors. This involves the processing of the visual data into ideal movements to intercept the ball. We have attempted to mitigate this risk by acquiring an extremely fast computer to perform the necessary calculations in real-time. However, there is still a smaller but present chance that the latency of each different element adds up to a point where the signals are not being sent to the motors in time.
- The result of processing time being too slow is again a lack of realistic play. However, this time it is not realistic to simply replace components to improve the speed of this system.
- The scope of the prediction algorithm can be simplified to speed up processing if required. Ultimately it can be reduced to simply moving the rod to intercept the ball. Similarly to the previous risk, if there is no feasible way to achieve realistic processing time, slow gameplay can be enforced.

8.0 Conclusion

In summary, the development of an automated foosball opponent addresses the absence of digital and interactive features in traditional foosball tables. By integrating real-time computer vision, mechatronic controls, and software–hardware coordination, this project will transform a familiar recreational activity into an engaging platform for both entertainment and engineering demonstration, not only for advanced players, but also beginners interested in the game.

On top of a new form of recreational play, the project provides a meaningful opportunity to showcase engineering skills across software and hardware domains, including control systems, computer vision, and embedded systems. The successful completion of this project will create a final product that is both accessible to a wide audience and valuable as a showcase of engineering in a recreational field.

9.0 References

[1] M. Rober. (2017, 03) Automatic bullseye, moving dartboard. Accessed: 01-Oct-2025. [Online]. Available: https://www.youtube.com/watch?v=MHTizZ_XcUM

[2] R. Janssen, J. Best, and M. Molengraft, "Real-time ball tracking in a semi-automated foosball table," 06 2009, pp. 128–139.

[3] J. D. Gutierrez-Franco, J. Inlow, J. Graham, and L. Huang, "Automated foosball table year 1: Final project report," California Polytechnic State University, Tech. Rep., Dec. 2013. [Online]. Available: https://digitalcommons.calpoly.edu/mesp/196

[4] D. L. Woods, J. M. Wyma, E. W. Yund, T. J. Herron, and B. Reed, "Factors influencing the latency of simple reaction time," *Frontiers in Human Neuroscience*, vol. 9, p. 131, 2015.

[5] StepperOnline. Nema 23 integrated easy servo motor 180w 3000rpm 0.6nm (84.98oz.in) 20-50vdc servo motor - short shaft. Accessed: 18-Oct-2025. [Online]. Available: https://www.stepperonline.ca/nema-23-integrated-easy-servo-motor-180w-3000rpm-0-6nm-84-98oz-in-20-50vdc-servo-motor-short-shaft-isv57t-180s.html

10.0 Appendices

Additional relevant research information and calculations are listed in their respective appendix sections below.

10.1 Appendix A: Gantt Chart

Figure 2 shows the projected timeline of tasks, key deliverables, and major milestones displayed in a Gantt chart.

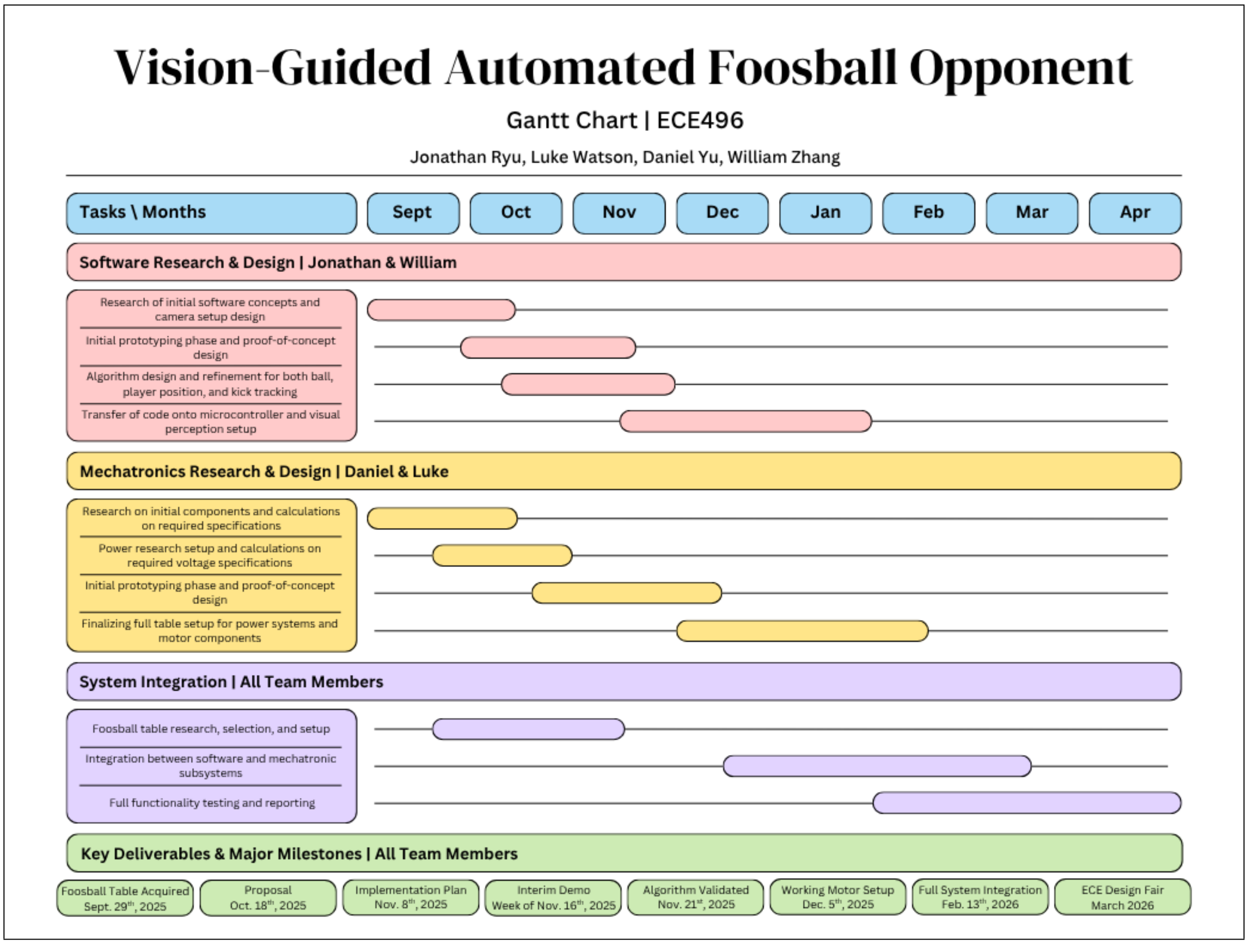
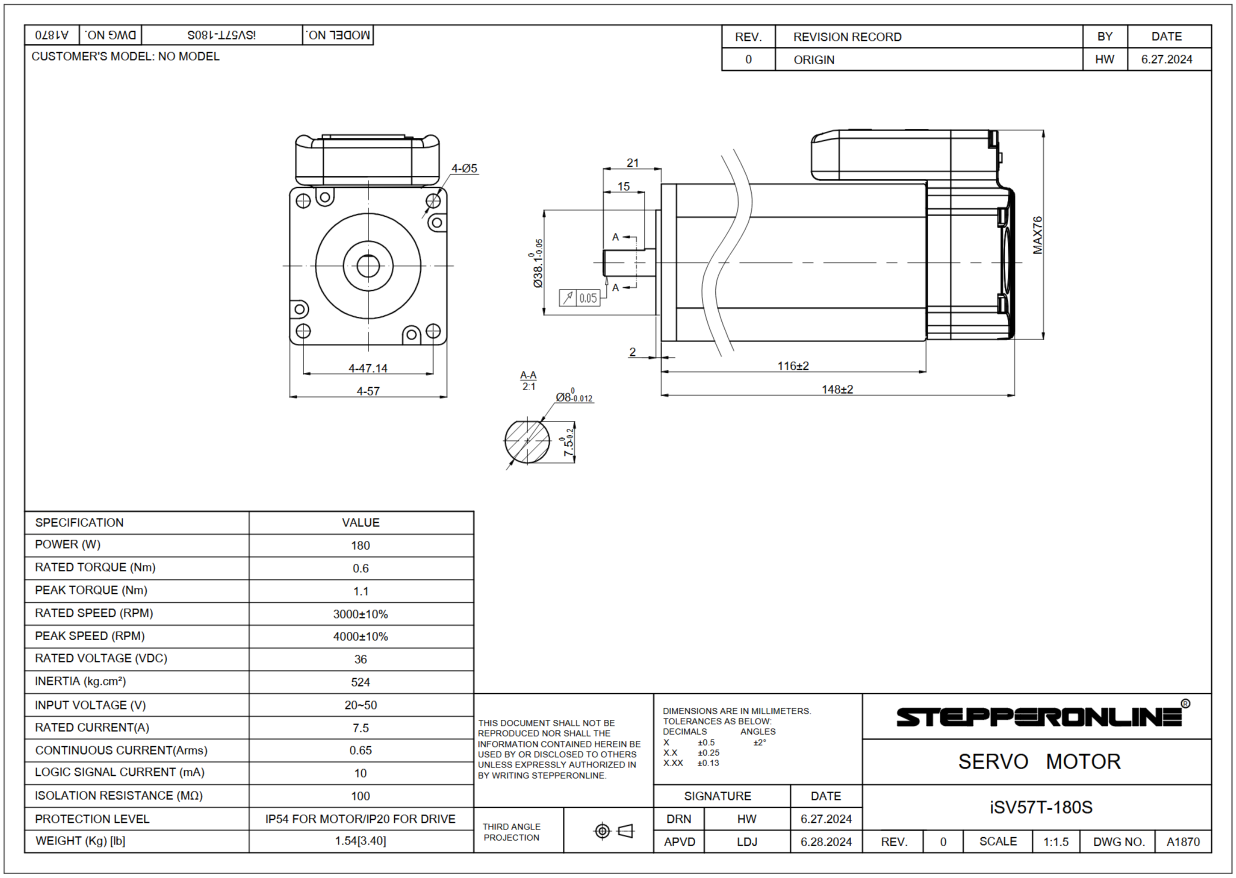


Figure 2. Gantt chart timeline for the 2025-2026 academic year.

10.2 Appendix B: Selected Motor Power Requirements

Figure 3 shows the specifications sheet for the selected motor.



SPECIFICATION	VALUE
POWER (W)	180
RATED TORQUE (Nm)	0.6
PEAK TORQUE (Nm)	1.1
RATED SPEED (RPM)	3000±10%
PEAK SPEED (RPM)	4000±10%
RATED VOLTAGE (VDC)	36
INERTIA (kg.cm²)	524
INPUT VOLTAGE (V)	20~50
RATED CURRENT(A)	7.5
CONTINUOUS CURRENT(Arms)	0.65
LOGIC SIGNAL CURRENT (mA)	10
ISOLATION RESISTANCE (MΩ)	100
PROTECTION LEVEL	IP54 FOR MOTOR/IP20 FOR DRIVE
WEIGHT (Kg) [lb]	1.54[3.40]

Figure 3. Selected motor datasheet [5].

10.3 Appendix C: Rotational Motor Specification Calculations

This appendix section shows the calculations behind the system requirement specifications for the rotational motor's speed and torque.

10.3.1 Rotational Speed Calculations

Assume the tangential speed of the foosman (attached to the rod) directly translates to the speed of the ball.

Most component specifications for speed are given in rotations per minute (RPM). To derive this from speed, this formula can be used:

$$\text{RPM} = \frac{\omega \cdot 60}{2\pi}$$

where ω is the angular velocity given by $\omega = \frac{v}{r}$.

The radius in this case is the distance from the center of the rod to the foot of the foosmen, since that is the point of contact with the ball.

The height of the center of the rod from the playing field is about 3”, or 76mm. On a professional table, the foosmen have a clearance of about 6.35mm. In practice this value can be much higher, and since a higher clearance will give a smaller radius, and therefore larger RPM, take a higher bound of 16mm. Therefore we can assume the radius to be approximately 60mm. Now we can graph this equation to see what RPM will be required for desired speeds.

$$y = \frac{v \cdot 60}{2\pi \cdot 0.06}$$

Thus to acheive a speed of 6m/s, around 1000RPM is required.

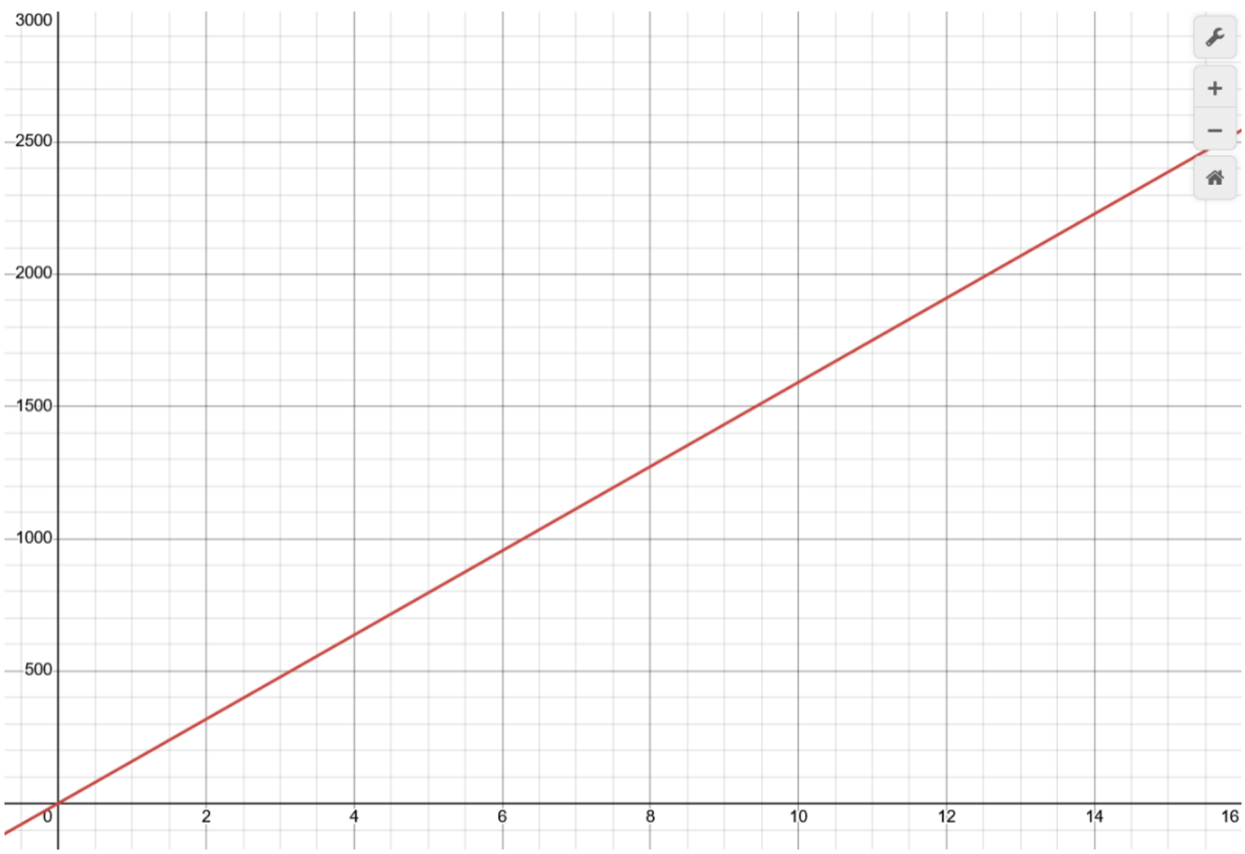


Figure 4. Speed to RPM.

10.3.2 Rotational Torque Calculations

The torque required to accelerate the rod to a desired velocity will be calculated using the formula:

$$\tau = I \cdot \alpha$$

where I is the total inertia of the system and α is the angular acceleration.

I can be calculated as the sum of the load inertia and the motor's rated inertia:

$$I = I_{load} + I_{rotor}$$

where load inertia is the combination of inertia from the rod and from the foosmen:

$$I_{load} = I_{rod} + I_{foosmen} = \frac{1}{2}m \cdot r^2 + \sum_{i=0}^{N} m_i \cdot r_i^2$$

Assume the mass of the rod to be 0.7kg[1] and the radius to be 8 mm[2]. This gives

$$I_{rod} = \frac{1}{2}(0.7)(0.008)^2 = 2.24 \cdot 10^{-5}\text{kg} \cdot \text{m}^2$$

Since we are focusing on two defensive lines of play, the most foosmen a rod can have is three, so we will use that value for N. Take a weight of 50g[3] and radius of 60mm, we get

$$I_{foosmen} = \sum_{i=0}^{N} m_i r_i^2 = N \cdot m \cdot r^2 = 3 \cdot 0.05 \cdot 0.06^2 = 5.4 \cdot 10^{-4}\ \text{kg} \cdot \text{m}^2$$

and subsequently

$$I_{load} = 2.24 \cdot 10^{-5} + 5.4 \cdot 10^{-4} = 5.625 \cdot 10^{-4}\text{kg} \cdot \text{m}^2$$

The reference datasheet in Appendix B shows the rotor inertia to be 524 $g \cdot cm^2$, giving

$$I = 5.265 \cdot 10^{-4} + 524 \cdot 10^{-7} = 5.79 \cdot 10^{-4}\ \text{kg} \cdot \text{m}^2$$

Next, to calculate angular acceleration, we first take a target speed of 6m/s and foosmen contact radius of 0.06m to calculate angular velocity:

$$\omega = \frac{v}{r} = \frac{6}{0.06} = 100\ \text{rad/s}$$

Time is more of a limiting factor than angle due to the fact that a servo should be able to rotate the rod a full 360 degrees, so we will use the

[1]Assuming a steel hollow rod.
[2]Using a typical rod with diameter of 5/8".
[3]Typical weight for a non-counterbalanced foosmen.

$$\alpha = \frac{\omega}{t} \text{ instead of } \alpha = \frac{\omega^2}{2\theta}.$$

Finally, with a 50ms timeframe for the motor to reach this desired speed from rest ($\omega_0 = 0$), we get

$$\alpha = \frac{\omega}{t} = \frac{100}{0.05} = 2000 \text{ rad/s}^2$$

Final torque is then calculated as:

$$\tau = 5.79 \cdot 10^{-4} \cdot 2000 = \ 1.158 \text{ N} \cdot \text{m}$$

10.4 Appendix D: Translational Motor Specification Calculations

The goal for the translational motor is to achieve realistic gameplay in a casual setting against a human opponent. We are aiming for a 1m/s translational speed with a reaction time of 0.05s[4]. In addition, since the translational motor needs to carry rotational motor, we can assume that it needs to carry at least 2kg of weight[5].

In order to determine the minimum required torque and speed, we first need to determine the total amount of force applied on this system using both the acceleration and force equations.

$$F_g = mg = 2\,\text{kg} \times 9.81\,\tfrac{\text{m}}{\text{s}^2} \approx 20\,\text{N}$$

$$a = \frac{v}{t} = \frac{1\,\frac{\text{m}}{\text{s}}}{0.05\,\text{s}} = 20\,\tfrac{\text{m}}{\text{s}^2}$$

$$F_a = ma = 2\,\text{kg} \times 20\,\tfrac{\text{m}}{\text{s}^2} = 40\,\text{N}$$

$$F_t = F_g + F_a = 20\,\text{N} + 40\,\text{N} = 60\,\text{N}$$

From our calculations above, we need to exert at least 40 Newtons of force. To determine the corresponding peak torque and speed requirements, we can apply the torque, angular velocity, and RPM formulas below.

$$\tau = rF\sin\theta = 0.02\,\text{m} \times 60\,\text{N} = 1.2\,\text{N}\cdot\text{m}$$

[4]Based on the same desired timeframe from standstill to full speed in Appendix C.

[5]From Appendix B, the selected motor is 1.54kg in weight and so we can assume 2kg as a safe margin of error with regards to additional miscellaneous weights such as screws.

The sine theta angle above is 90 degrees since the linear guide rail creates a constant perpendicular angle, resulting in the sine component of the torque equation to be equal to 1.

$$\omega = \frac{v}{r} = \frac{10\,\frac{\text{m}}{\text{s}}}{0.02\,\text{m}} = 50\,\frac{\text{rad}}{\text{s}}$$

The radius component in both the torque and angular velocity equations is estimated to be around 20mm from research completed on current off-the-shelf linear guide rail models.

$$\text{RPM} = \frac{\omega \times 60}{2\pi} = \frac{50 \times 60}{2\pi} \approx 500\,\text{RPM}$$

Based on our calculations, the required peak torque of our motor is 1.2Nm while the speed of the motors to maintain our goal of 1m/s needs to be at least capable of 500RPM. If needed, a gearbox can be added to this system to further increase the output torque.